

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	S.Roshni	Reg. No.:	192525185
Section / Batch:	AI&ML	Date:	10/07/2026

Code of Conduct

I S.Roshni(192525185)_____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Roshni_____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
GRAND TOTAL:			100 Marks

CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1

C Code – Array Traversal
10 marks



Q2

C Code – Linear Search
10 marks



Q3

C Code – Binary Search Mid
10 marks



Q4

C Code – Pointer with Array
10 marks



Q5

C Code – Array Update
10 marks



Q1 C Code – Array Traversal

```
#include <stdio.h>

int main() {
    int arr[] = {5, 10, 15, 20};
    int i, sum = 0;
    for(i = 0; i < 4; i++)
        sum += arr[i];
    printf("%d", sum);
    return 0;
}
```

Your Answer

50

CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1

C Code – Array Traversal
10 marks



Q2

C Code – Linear Search
10 marks



Q3

C Code – Binary Search Mid
10 marks



Q4

C Code – Pointer with Array
10 marks



Q5

C Code – Array Update
10 marks



Q6

C Code - Linked List Node
10 marks



Q2 C Code – Linear Search

```
#include <stdio.h>

int main() {
    int arr[] = {2,4,6,8,10};
    int key = 8;
    int i;
    for(i=0;i<5;i++){
        if(arr[i]==key){
            printf("%d",i);
            break;
        }
    }
}
```

Your Answer

3



CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1

C Code – Array Traversal
10 marks



Q2

C Code – Linear Search
10 marks



Q3

C Code – Binary Search Mid
10 marks



Q4

C Code – Pointer with Array
10 marks



Q5

C Code – Array Update
10 marks



Q6

C Code – Linked List Node
10 marks



Q3 C Code – Binary Search Mid

```
#include<stdio.h>
int main(){
int low=0,high=7;
int mid;
mid=(low+high)/2;
printf("%d",mid);
}
```

Your Answer

3



CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1

C Code – Array Traversal
10 marks



Q2

C Code – Linear Search
10 marks



Q3

C Code – Binary Search Mid
10 marks



Q4

C Code – Pointer with Array
10 marks



Q5

C Code – Array Update
10 marks



Q6

C Code – Linked List Node
10 marks



Q5 C Code – Array Update

```
#include<stdio.h>
int main(){
int arr[]={1,2,3};
arr[1]=arr[0]+arr[2];
printf("%d",arr[1]);
}
```

Your Answer

4

 Save Answer

CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1

C Code – Array Traversal
10 marks



Q2

C Code – Linear Search
10 marks



Q3

C Code – Binary Search Mid
10 marks



Q4

C Code – Pointer with Array
10 marks



Q5

C Code – Array Update
10 marks



Q4 C Code – Pointer with Array

```
#include<stdio.h>

int main(){
    int arr[]={10,20,30};
    int *p=arr;
    printf("%d",*(p+2));
}
```

Your Answer

30



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CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1

C Code – Array Traversal
10 marks



Q2

C Code – Linear Search
10 marks



Q3

C Code – Binary Search Mid
10 marks



Q4

C Code – Pointer with Array
10 marks



Q5

C Code – Array Update
10 marks



Q6

C Code - Linked List Node
10 marks



Q7 C Code - Time Complexity Loop

```
for(i=0;i<n;i++)
    printf("*");
The complexity is
```

Your Answer

$O(n)$

 Save Answer

CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1

C Code – Array Traversal
10 marks



Q2

C Code – Linear Search
10 marks



Q3

C Code – Binary Search Mid
10 marks



Q4

C Code – Pointer with Array
10 marks



Q5

C Code – Array Update
10 marks



Q6

C Code - Linked List Node
10 marks



Q8 C Code - Nested Loop

```
for(i=0;i<n;i++)  
for(j=0;j<n;j++)  
printf("*");
```

Your Answer

$O(n^2)$

 Save Answer

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CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1

C Code – Array Traversal
10 marks



Q2

C Code – Linear Search
10 marks



Q3

C Code – Binary Search Mid
10 marks



Q4

C Code – Pointer with Array
10 marks



Q5

C Code – Array Update
10 marks



Q6

C Code - Linked List Node
10 marks



Q9 C Code - Binary Search Complexity

```
while(low<=high){  
mid=(low+high)/2;  
}
```

Your Answer

$O(\log n)$

 Save Answer

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CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q9 C Code - Binary Search Complexity

```
while(low<=high){  
    mid=(low+high)/2;  
}
```

Your Answer

o(log n)

Save Answer

 **SIMATS**
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SIMATS Online Programming Lab
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CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code - Linked List Node
10 marks

Q7

Q10 C Code - Pointer Increment

```
#include<stdio.h>  
  
int main(){  
    int a[]={5,10,15};  
    int *p=a;  
    p++;  
    printf("%d",*p);  
}
```

Your Answer

10

Save Answer

CO1_AT1_C CODE OUTPUT PREDICTION

QUESTIONS

Q1

C Code – Array Traversal

10 marks



Q2

C Code – Linear Search

10 marks



Q3

C Code – Binary Search Mid

10 marks



Q4

C Code – Pointer with Array

10 marks



Q5

C Code – Array Update

10 marks



Q6

C Code - Linked List Node

10 marks



Q6 C Code - Linked List Node

```
#include<stdio.h>

struct Node{
int data;
struct Node *next;
};

int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Your Answer

20